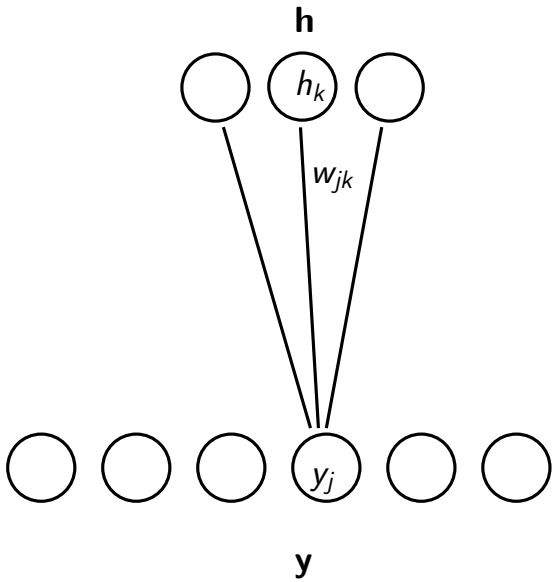
# Learning the allowable transformations of simple shapes using 3-way interactions

Roland Memisevic, Geoffrey Hinton

University of Toronto

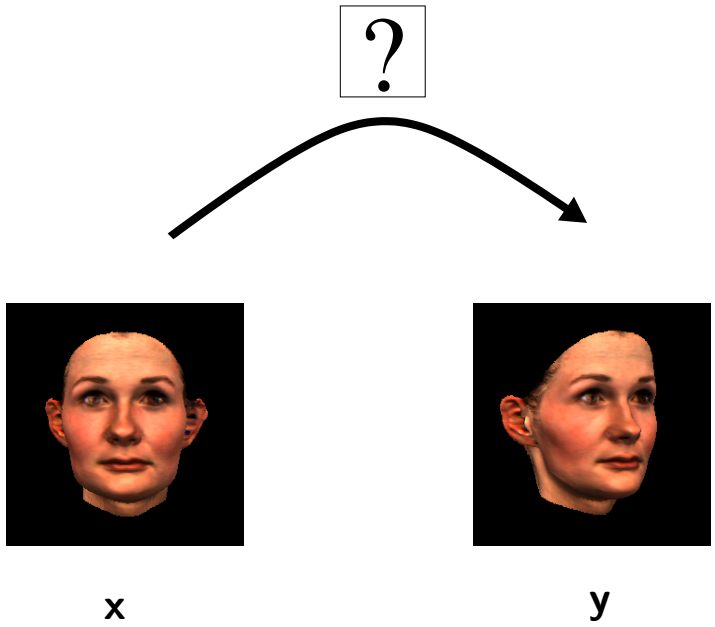
Dec 1, 2007

# 'Natural images are not random'



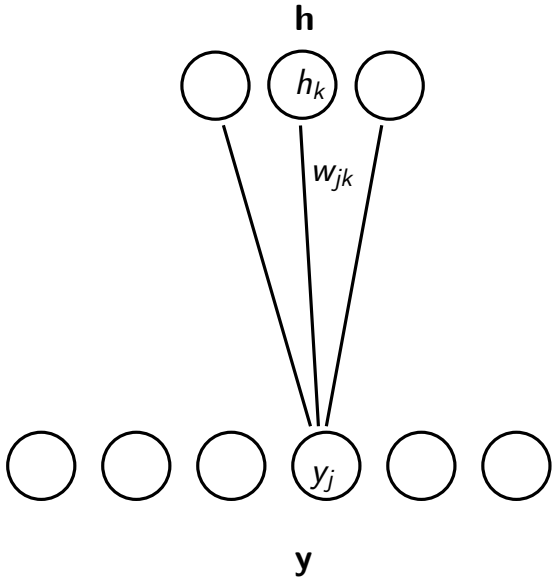

- A common way to capture the statistics of natural images is by learning *filters* from patches: ICA, PCA, NMF, ...
- Bi-partite models explain pixels $y_i$ using hidden "causes" $h_k$.
- What about *transformations* of natural images?
- Can we learn how images *change*?

# Learning data *transformations*




- How can we learn *transformations* of data...
- **...and keep the statistics of the data in mind?**
- Idea: *Condition* on the input image.
- Hidden units now become 'mapping' units that learn to encode transformations.

# Restricted Boltzmann machines




- (Think of **y** and **h** as binary.)
- One way to learn (static) filters is by using the *energy*:

$$E(\mathbf{y}, \mathbf{h}) = -\sum_{jk} w_{jk} y_j h_k$$

- ... and defining:

$$p(\mathbf{y}, \mathbf{h}) = \frac{1}{Z} \exp(-E(\mathbf{y}, \mathbf{h}))$$

# Mapping units

- To condition, make the weights a function of the inputs. For example:

$$w_{jk}(\mathbf{x}) = \sum_i w_{ijk} x_i$$

- We get:

$$E(\mathbf{y}, \mathbf{h}; \mathbf{x}) = -\sum_{ijk} w_{ijk} x_i y_j h_k$$

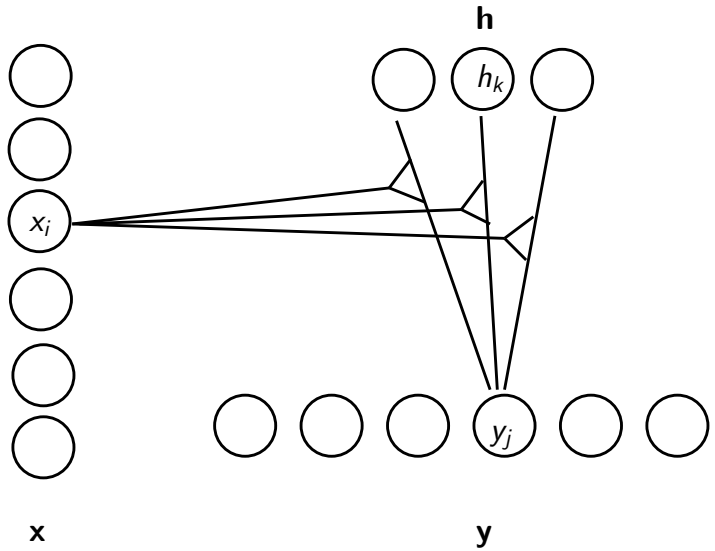
- Now model the **joint conditional** over data and hiddens as

$$p(\mathbf{y}, \mathbf{h}|\mathbf{x}) = \frac{1}{Z(\mathbf{x})} \exp(-E(\mathbf{y}, \mathbf{h}; \mathbf{x}))$$

$$Z(\mathbf{x}) = \sum_{\mathbf{y},\mathbf{h}} \exp(-E(\mathbf{y}, \mathbf{h}; \mathbf{x}))$$

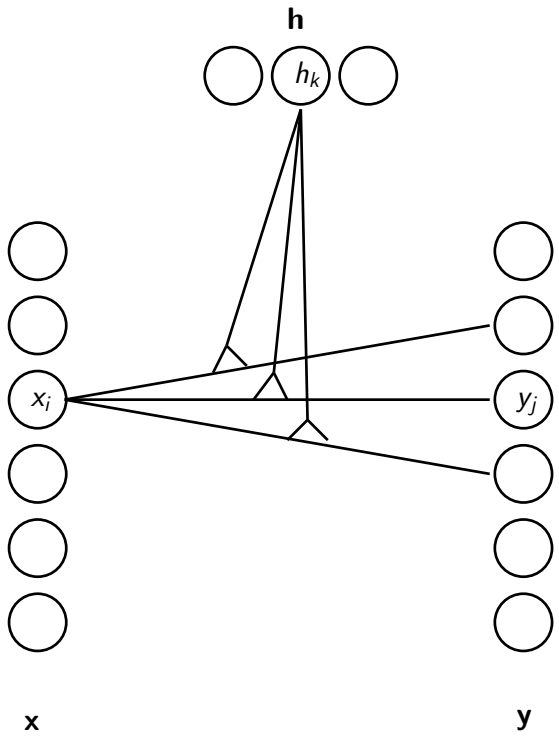
- (Hinton, Lang 1985), (Sejnowski, 1986), (He, Zemel 2004).

# Two views: 1. Input-dependent filters




- Conditional RBM. *Potentials* depend on inputs:
- A CRF, that *learns* its features.
- Input-dependent filters.

# Two views: 2. Modulated regression




- Modulated regression. Functions, modulated by hidden units.
- Exponential mixture of experts (with weight sharing).
- Mapping units provide a *factorial* code for transformations.

# Training and inference

## Training

- For training, maximize $L(W) = \sum_\alpha \log p(\mathbf{y}^\alpha | \mathbf{x}^\alpha)$
- Gradient:

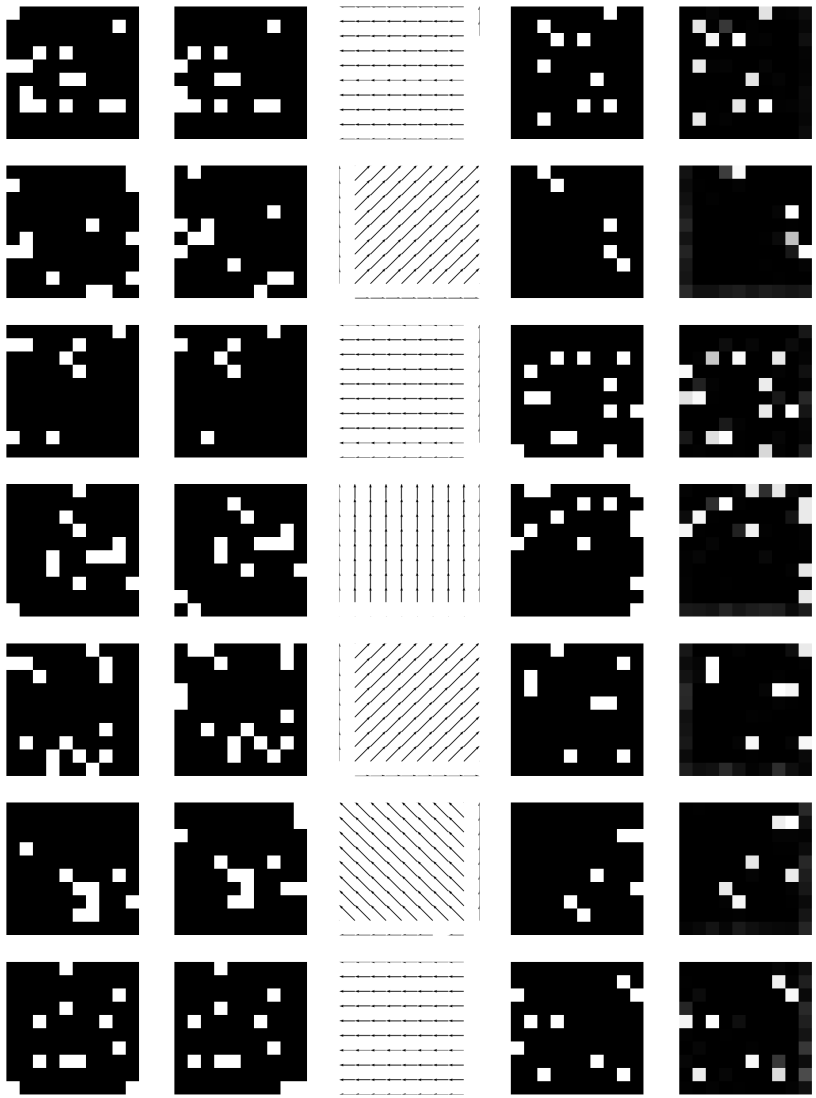
$$\frac{\partial L}{\partial W} = \sum_\alpha \left[ \sum_{\mathbf{h}} p(\mathbf{h}|\mathbf{y}^{\boldsymbol{\alpha}}, \mathbf{x}^{\boldsymbol{\alpha}}) \frac{\partial E(\mathbf{h}, \mathbf{y}^{\boldsymbol{\alpha}})}{\partial W} - \sum_{\mathbf{h},\mathbf{y}} p(\mathbf{y}, \mathbf{h}|\mathbf{x}^{\boldsymbol{\alpha}}) \frac{\partial E(\mathbf{h}, \mathbf{y})}{\partial W} \right]$$

- Contrastive divergence.

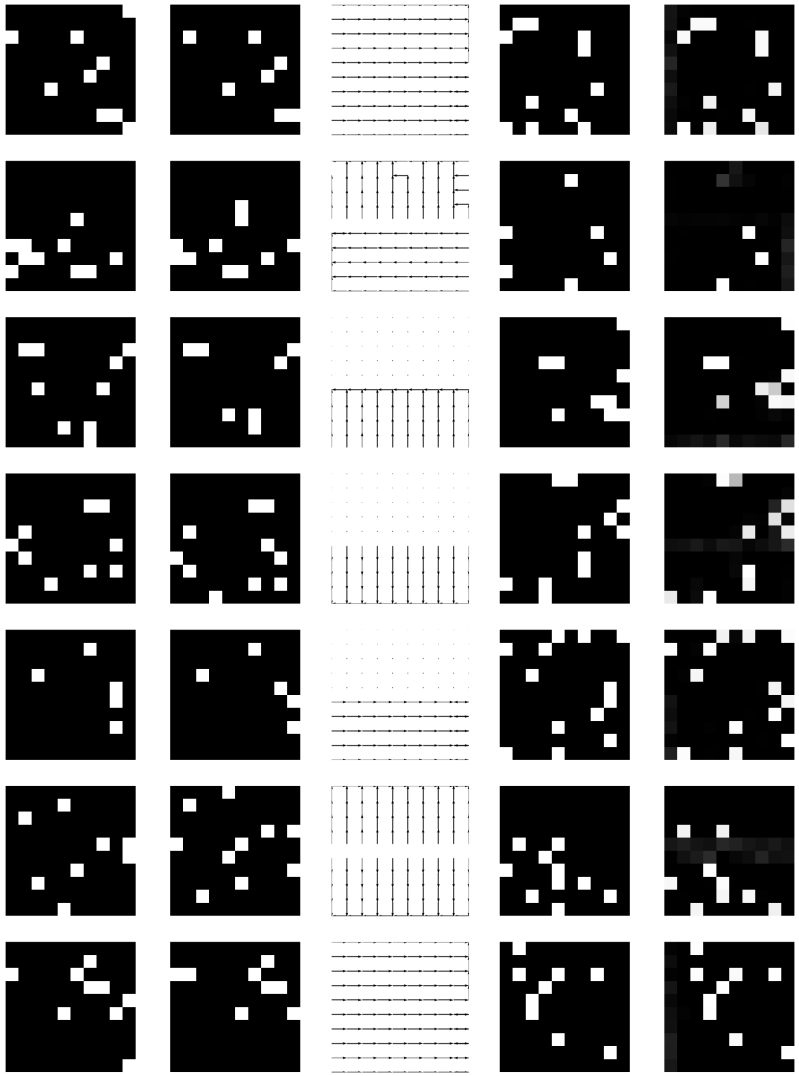
## Inference

- To infer a transformation, compute $p(\mathbf{h}|\mathbf{y}, \mathbf{x})$. Easy, because of conditional independencies.
- Optical flow represented *implicitly* in the model (as a binary vector). But we can visualize what the model 'thinks' by plotting where each pixel wants to go *the most*:

# Example



# Factorial flowfields

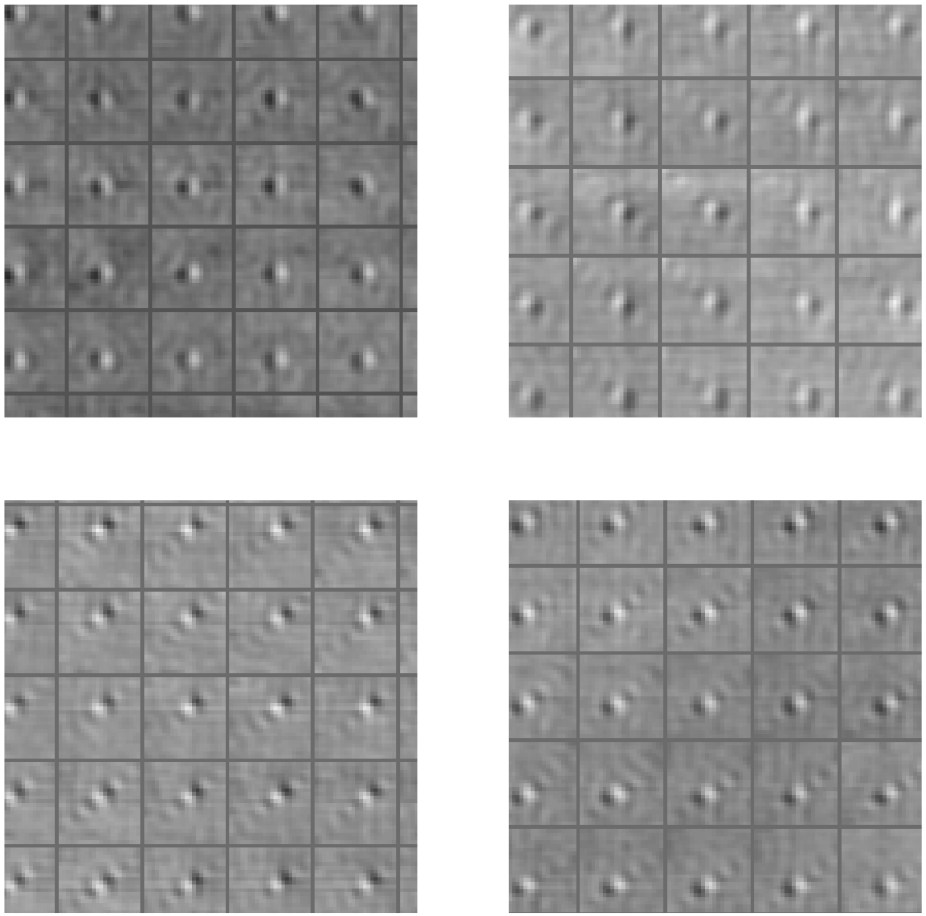


# Exponential family

- Instead of using binary units, it is possible to use any distribution from the **exponential family** as the conditionals for **h** or **y**.
- Conditional analog of (Welling, et al. 2005).
- Can use convolutional version for large images.
- Can model nonlinear transformations: $w_{jk}(\mathbf{x}) = f_{jk}(\mathbf{x})$, and do feature extraction first

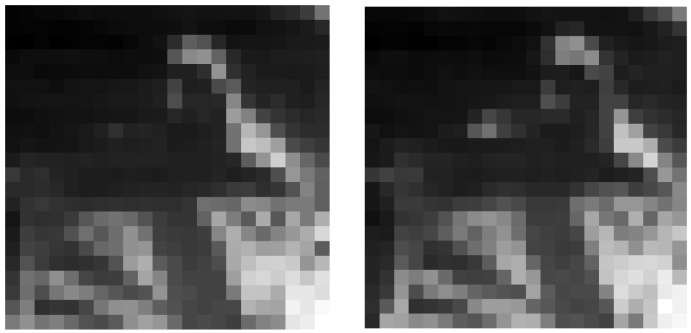
# Flowfields on natural images

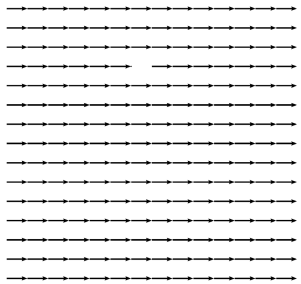
- Receptive fields become *conditional receptive fields*.
- Learning on few hours of television broadcast (van Hateren):



# Modeling "optical flow"

- Hidden units learn *correlation patterns* in **x** and **y**.
- Spatial pooling facilitates generalization and noise suppression:





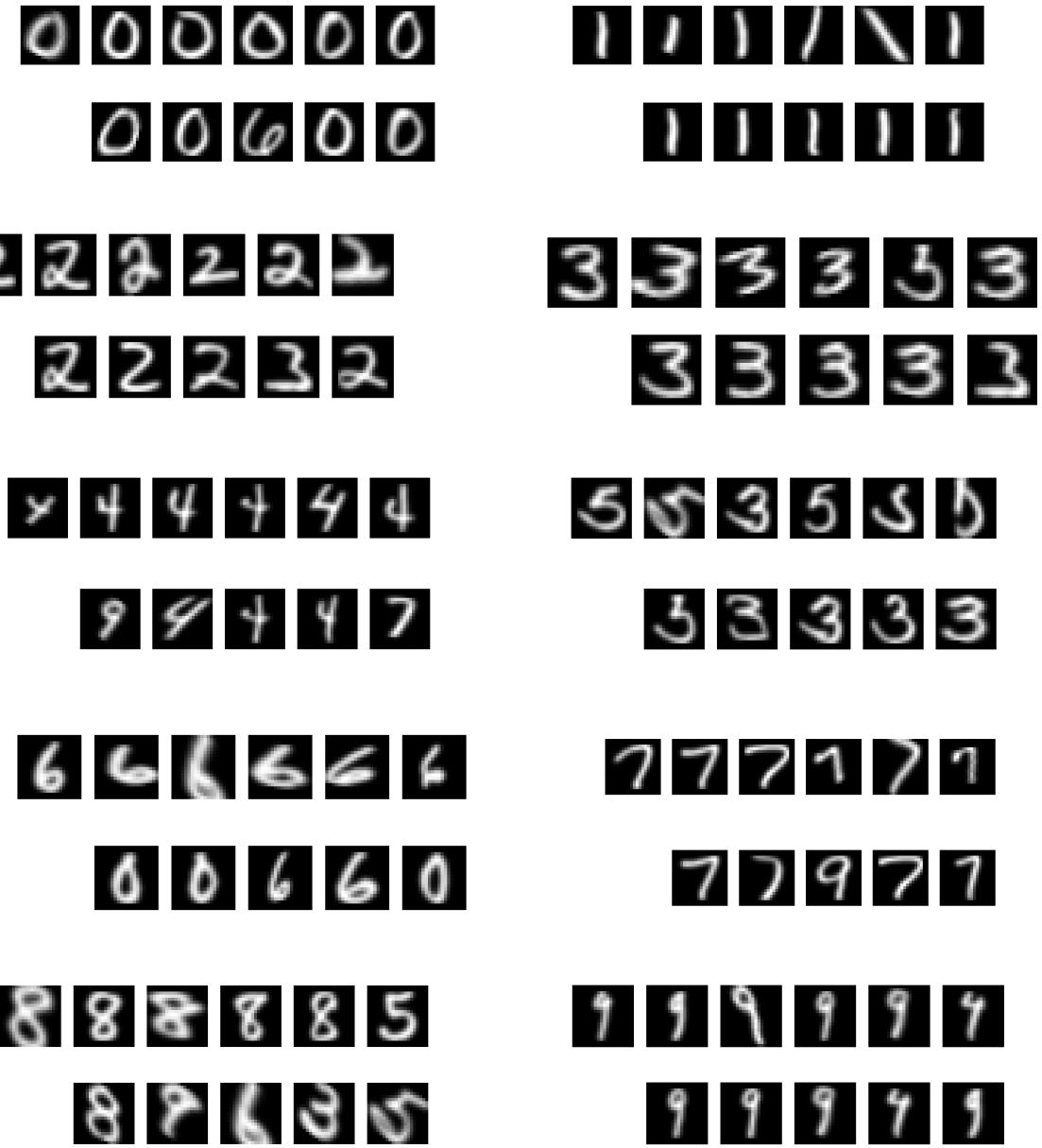
# Learning an invariant metric

- We are learning a non-linear manifold *implicitly*: As the set of transformations that generate it.
- After training the model, we can define an invariant metric as follows:

"Transformation metric":

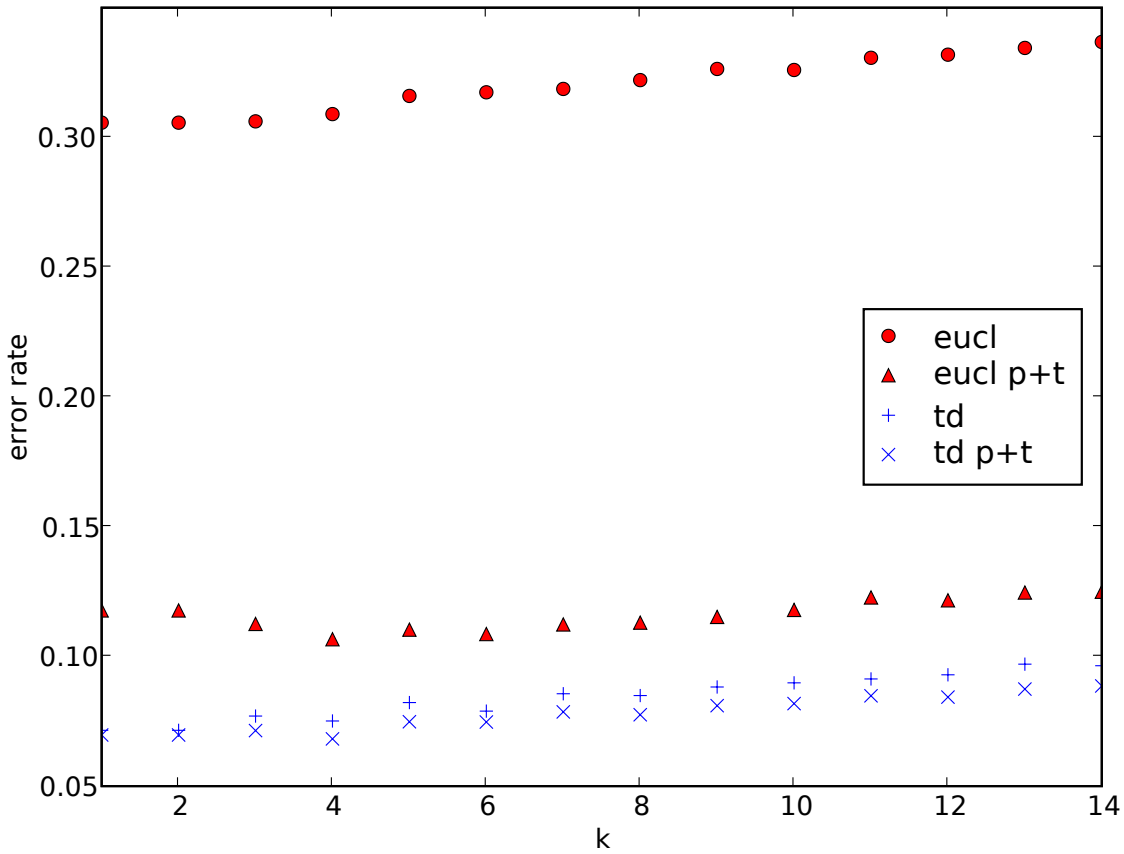
- To measure the dissimilarity $d(\mathbf{x}, \mathbf{y})$ between new cases $\mathbf{x}$ and $\mathbf{y}$:
  1. Set $\hat{\mathbf{h}} = \arg\max_{\mathbf{h}} p(\mathbf{h}|\mathbf{x}, \mathbf{y})$
  2. Set $\hat{\mathbf{y}} = \arg\max_{\mathbf{y}} p(\mathbf{y}|\mathbf{x}, \hat{\mathbf{h}})$
  3. Set $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{y} - \hat{\mathbf{y}}\|$
- One-step-reconstruction error. ("CD at test-time")
- No knowledge included!

# Some nearest neighbors



# k-nearest neighbors on affine digits

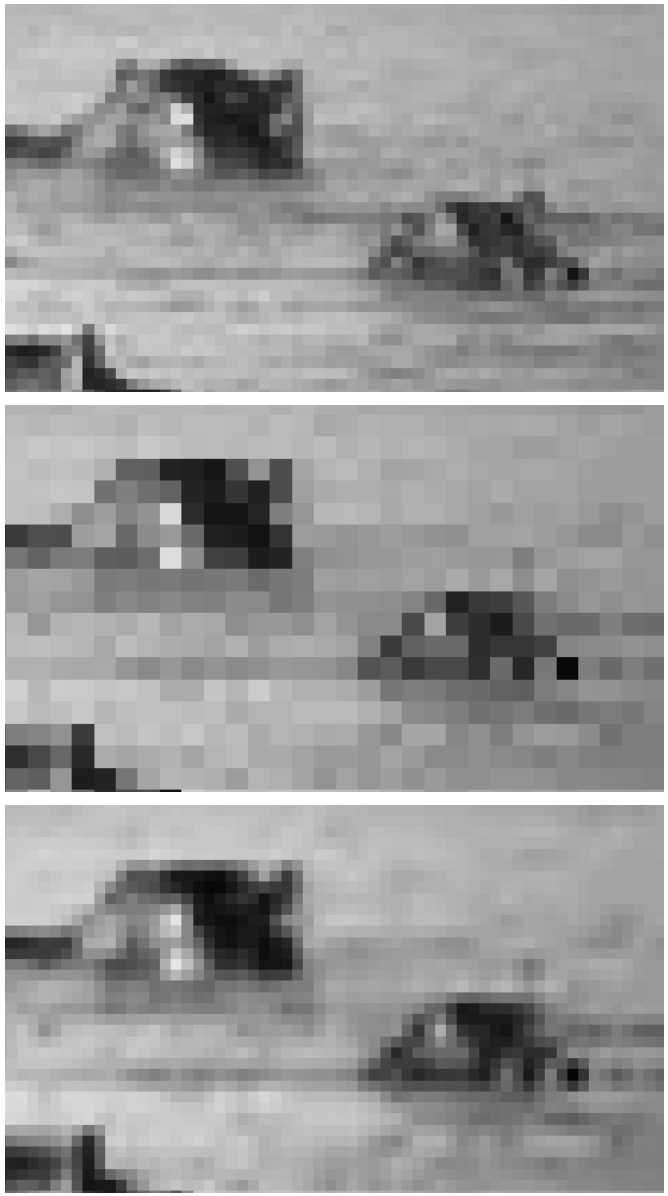
- After learning a metric, we can use it, for example, to do classification.



# Super-resolution



# Super-resolution



# Discriminative de-noising

- If we know how to *corrupt* images, we can learn how to de-corrupt them.



# Discussion and further applications

- The task of modeling transformations is inherently "three-way".
- Here, unlike bi-linear models, only one group of latent variables.
- Learning/inference therefore straightforward.
- Biology?
- Other potential applications: Transfer learning, Learning to cluster, Temporal de-noising, Video compression, Stereo, Depth...